

the expanse of the bigger cities connected by the nation's arterial highway network. Despite an electrifying future in India's EV space, we still wouldn't recommend holding your breath on Tesla's electric cars launching in India anytime soon. Don't be surprised to see an increased number of electric bicycles on the road as people's daily commuter transport option of choice going forward, though.

India's space endeavours in 2023

Last year, ISRO's failure to launch the SSLV (Small Satellite Launch Vehicle) managed to grab all the headlines in India. But that doesn't mean India's space-tech needle didn't move forward. Oh, no sir!

Bengaluru-based Pixxel became India's first private startup to launch its satellite, Shakuntala, into a Low-Earth Orbit (LEO) on a SpaceX Falcon 9 rocket in April 2022. Pixxel consolidated its LEO imaging network by launching another satellite through ISRO's Polar



Satellite Launch Vehicle (PSLV)'s C54 mission. As far as 2023 is concerned, according to Awais Ahmed, CEO of Pixxel, the Indian space-tech startup aims to start beaming down imaging data for customers on Earth. Not just that, Pixxel also wants to establish its own satellite manufacturing and assembly factory in India within 2023 – this move will definitely help broaden the scope of space-tech missions from India in the near future.

Apart from Pixxel, Skyroot Aerospace is another very interesting Indian space-tech startup that's pushing the boundaries of what's possible. Skyroot is India's first startup to launch a private rocket–Vikram S–to reach sub-orbital space.

Where Skyroot's championing the cause of made-in-India rockets, another space-tech startup called Agnikul is expected to launch India's first private space mission that actually reaches orbital space, hopefully within 2023 itself, setting itself an audacious goal of at least two satellite launches every month in the near future.

Both these startups are expected to break new ground and reach new milestones, as far as private Indian space endeavours are concerned in 2023, rivalling the likes of France's Arianespace and America's ULA space companies.

There are several Indian space startups that could come-of-age in 2023, finding their groove and mastering the incredibly tough-to-conquer final frontier of space missions. ISRO could announce a slew of milestones, including but not limited to the country's first-manned mission of Gaganyaan. ISRO's Chandrayaan-3 is expected to launch in June 2023, hoping to succeed where Chandrayaan-2 failed, by landing a rover, orbiter and lander on the moon. ISRO is also launching 'XPoSat', a space observatory to uncover the hidden mysteries of cosmic X-rays – the mission will be launched on ISRO's SSLV rocket.

India's climate tech horizon in 2023

One of the biggest challenges faced by humankind, climate action is of paramount importance, and India isn't shying away from its responsibilities – across the board from governmental agencies to private entrepreneurs, academia to consumers, everyone understands the importance of positive, sustainable climate action going forward.

Venture capital isn't shying away from backing climate tech startups in India, as \$7bn was raised by Indian Climate Tech related start-ups in 2021, according to a Unitus Capital report. With over ₹2 lakh crore in planned

investments in India for using sunlight, water, and air to produce energy, boosting renewable energy production and consumption remains a big ticket item for India. It is expected that by 2040, around 49% of the total electricity will be generated by renewable energy as



more efficient batteries will be used to store electricity, which will further cut the solar energy cost by 66% as compared to the current cost. The country has set an ambitious goal of generating 500 Gigawatt of renewable energy by 2030, achieved by EV adoption, green hydrogen energy, solar energy expansion and investments in energy storage solutions. Expect to see new and old startups in these arenas to create waves all throughout 2023 and beyond.

All this continued investment in clean energy will reduce pollution levels as villages become self-sustainable with their use of clean energy. But experts believe that breakthroughs in quantum computers, space mining and advanced artificial intelligence engines might have a shot at solving some of the fundamental problems of climate change that we just cannot confront fully right now. From long term, complex ecosystem impacts we haven't even begun to fathom to our slow detachment from fossil fuels, it's clear that climate action needs more innovative solutions, and constant ingenuity of our best scientists, mathematicians, AI and data scientists. It's novel tools and technologies to understand and solve this immensely dynamic challenge. **[d]**